

NOETHER: Data and Artifact Availability Statement

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ACM Reference Format:

Meng Li, Xiaohua Yang, Jie Liu, and Shiyu Yan. 2026. NOETHER: Data and Artifact Availability Statement. *ACM Trans. Softw. Eng. Methodol.* 0, 0, Article 0 (2026), 2 pages. <https://doi.org/XXXXXXX.XXXXXXX>

Data and Artifact Availability

To support independent verification, the supplementary code and data are archived on Zenodo under DOI <https://doi.org/10.5281/zenodo.20250634>. The same materials are supplied through the submission system with the same file manifest and content hash.

Review-stage supplementary archive. The following supplementary materials are submitted alongside the manuscript and are available through the submission system under SHA-256 content hash. The hash is computed over the concatenated tar-archive of the listed items and reported in the final manuscript at acceptance.

- **S1** Python reference implementation of CONSTRUCT-MP described in supplementary S9, including the Boltzmann and equivariant-ML instantiations.
- **S2** The full PWR-MR corpus, with each MR annotated by canonical structural component, NOETHER MetaPattern, and source equation.
- **S3** The SE(3)-equivariant point-cloud testing harness, including the mutation generators and MR sets used in the case-study comparison.

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2026. ACM 1557-7392/2026/0-ART0
<https://doi.org/XXXXXXX.XXXXXXX>

- **S4** A reproducibility manifest documenting the random seeds, model checkpoints, and dataset versions referenced in the case study.
- **S5** GenMorph/PIT pilot material, head-to-head summaries, parsed mutation results, and statistical scripts for the secondary executability and complementarity analyses.
- **S6** Query-optimiser pilot material for the relational-equivalence extension, including sampled Calcite/QED plan pairs and oracle scripts.
- **S7** Defects4J algebra-rich substrate material, pre-registration records, SUT-selection criteria, and $\mathcal{L}^*$ -blindness evidence.
- **S8** METRIC+ comparison material, Sun et al. subject re-implementations, scope analysis, and head-to-head protocol/results files.
- **S9** Appendices and detailed empirical tables migrated out of the Fast Impact main body for length conformance.
- **S10** Home-field generation-to-detection checks, rank/faithfulness scripts, and additional Invariance-Blindness sanity checks.
- **S11** Industrial N5 coverage classification inputs and results.
- **S12** Cross-domain N5 numerical-library and signal-processing derivation checks and outputs.

Public archival release. The public archival package is deposited on Zenodo under DOI <https://doi.org/10.5281/zenodo.20250634>. The camera-ready version will additionally record the final SHA-256 hash of the submission archive. We aim for the *Available* and *Functional* artefact-evaluation badges.

A Relocated figure and table material

This appendix collects figure and table material moved out of the main text to keep the Fast Impact body focused on the core evidence chain.

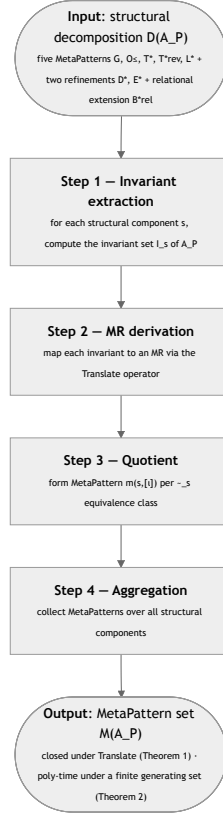


Fig. 4. CONSTRUCT-MP derives $\mathbb{M}(\mathcal{A}_P)$ through invariant extraction, translation, quotienting, and aggregation.

B Noether-style derivation of m_{adj} and Boltzmann specialisations

B.0.1 A Noether-style derivation of m_{adj} . The framework's name is methodological. Noether's first theorem replaces an empirically curated catalogue of conservation laws with a derivation from the symmetry structure of an action functional [40]. We do not invoke Noether's theorem as a theorem about programs (programs do not, in general, possess an action functional in the variational-calculus sense). What the framework does mirror is the *methodological move*: instead of cataloguing observed invariants, derive them from a structural source. We illustrate the move on m_{adj} .

Consider the bilinear form on solution-adjoint pairs

$$\mathcal{F}[\phi, \phi^\dagger] = \langle \phi^\dagger, B\phi \rangle - \langle \phi, B^\dagger \phi^\dagger \rangle, \quad (6)$$

where B is the Boltzmann operator and $B^\dagger$ its formal adjoint under the inner product $\langle f, g \rangle = \int f g d\Omega dE dr$. The bilinear form $\mathcal{F}$ is identically zero on solutions of the forward and adjoint

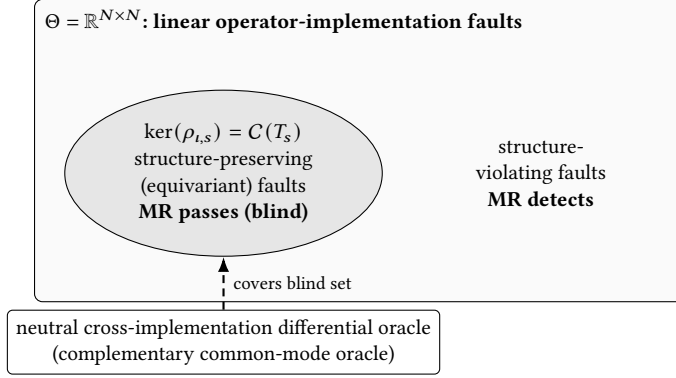


Fig. 5. The Invariance-Blindness Theorem separates structure-preserving faults from MR-detectable faults for G and T^* .

Table 8. Documented out-of-scope program-family classes name candidate extensions beyond the current decomposition.

Program-family class	Missing / non-algebraic structure	Candidate	MetaPat-	Empirical witness
tern				
<i>Candidate additional MetaPatterns</i>				
Symplectic systems	Symplectic 2-form, not self-adjointness	Symplectic-volume		N -body / celestial integrators
Sheaf-theoretic / categorical	Functors and natural transformations	Categorical-square commutativity		Certified compilers; type-driven transforms
Probabilistic / martingale	Conditional-expectation invariants $\mathbb{E}[f(\mathbf{x}_t) \mid \mathcal{F}_s]$	Martingale		Markov-chain Monte Carlo (MCMC) samplers; SGD analyses
Topological invariants	Homological signature	Topological; metric-stability M_{lip}		Audit orphan (§3.4.3)
Label-consistency (supervised ML)	Prediction-vs-ground-truth-label operator	Label-consistency		Case study cat-(i) wrong-sign, 0/5
Parameter-distribution divergence	$D_{\text{KL}}(p_\theta \ p_{\theta+\Delta\theta}) \leq \tau$ on parameter measures	Distribution-divergence		DeepCrime pilot cat-v-02/04/05 undetected
<i>Domain out-of-scope: no operator-algebraic representation</i>				
General web applications	State-machine / business-rule semantics	n/a		n/a
RLHF reward models / LLM agents	Statistical / measure-theoretic invariants	n/a		n/a
Distributed-consensus protocols	Conflict-free replicated data type (CRDT) / message-trace; linearisability	n/a		n/a
Compiler-internal optimisations	Refinement on operational semantics	n/a		n/a

Table 9. Representative PWR MRs are placed under canonical NOETHER symbols.

MR ID	Plain-text MR	Canonical symbol	Canonical class
Bur-Phy-01	Step-splitting invariance	G	$f_{\text{inv.con}}$
Bol-Phy-11	$\Sigma_a \uparrow \Rightarrow k_{\text{eff}} \downarrow$	$O_{\leq}$	$f_{\text{mono.stat}}$
Dif-Alg-01	Diamond-difference h^2 convergence	$\mathcal{L}^*$	$f_{\text{conv.lim}}$
Bur-Phy-08	Iodine pit qualitative shape	$O_{\leq} (\mathcal{D}^* \text{ refinement})$	$f_{\text{mono.shape}}$
Bur-Alg-04	CRAM no-worse-than transmutation-trajectory analysis (TTA)	$\mathcal{L}^* (\mathcal{E}^* \text{ refinement})$	$f_{\text{conv.rate}}$
Adj-Phy-01	Adjoint reciprocity	T^*	$f_{\text{adj.dual}}$
Rev-Phy-01	Collisionless reversibility	$\mathcal{T}_{\text{rev}}^*$	$f_{\text{rev.traj}}$

Table 10. The equivariant-ML case study compares Set N, LLM-prompted, and literature MR sets on four mutation categories.

	Set N (NOETHER)	Set L (LLM)	Set B (Lit.)
Detection rate	7/20	2/20	0/20
Structural coverage (coverage _{NOETHER})	1.00	0.20	0.20
Unique detections	5	0	0
Detected cat. (i) wrong-sign loss	0/5	0/5	0/5
Detected cat. (ii) equivariance break	2/5	2/5	0/5
Detected cat. (iii) precision degradation	0/5	0/5	0/5
Detected cat. (iv) gradient-reversal sign	5/5	0/5	0/5

equations: $B\phi = S$, $B^\dagger \phi^\dagger = S^\dagger$ with appropriate boundary conditions imply $\mathcal{F}[\phi, \phi^\dagger] = \langle \phi^\dagger, S \rangle - \langle \phi, S^\dagger \rangle$, and the standard reciprocity identity makes the right-hand side zero. The vanishing of $\mathcal{F}$ is a *conserved current* in the Noether sense: it is annihilated by the dual symmetry $\phi \leftrightarrow \phi^\dagger$, $B \leftrightarrow B^\dagger$. CONSTRUCT-MP's Step 1 extracts $\mathcal{F} = 0$ as the T^* -MetaPattern invariant; Translate converts it into the executable MR

$$\rho_{\text{adj}} : \quad |\langle \phi^\dagger, S \rangle - \langle \phi, S^\dagger \rangle| \leq \tau, \quad (7)$$

which the testing harness checks on a forward and an adjoint solver run with externally specified sources $S, S^\dagger$. The MR is therefore a Noether-style consequence of the duality symmetry rather than a manually catalogued reactor-physics property: the symmetry $\phi \leftrightarrow \phi^\dagger$ is the structural source, the bilinear form is the conserved current, and the MR is the conservation law in testable form. The same template applies to m_{rev} with the involution $\mathcal{T} : \phi(\mathbf{r}, \Omega, E, t) \mapsto \phi(\mathbf{r}, -\Omega, E, -t)$ as the symmetry. The template's general statement is: given a symmetry of the program family's operator algebra, CONSTRUCT-MP extracts the invariant of the symmetry and Translate converts it into an executable MR.

B.0.2 Specialisation to neutron transport, diffusion, and burnup. The Boltzmann formulation is the most general; specialised solvers are obtained by truncating or projecting $\mathcal{A}_{\text{Boltz}}$. Neutron transport solvers retain the full angular dependence; the same five MetaPatterns apply. Neutron diffusion solvers truncate angular dependence to its P_1 approximation: the $\mathcal{T}_{\text{rev}}^*$ MetaPattern contracts (diffusion is dissipative), removing m_{rev} . Burnup-coupled solvers introduce the Bateman ODEs; the Bateman operator e^{At} enriches multiple structural components (G semi-group, $O_{\leq}$ linearity, $\mathcal{D}^*$ overshoot/S-curve, $\mathcal{E}^*$ CRAM-vs-TTA bounds).

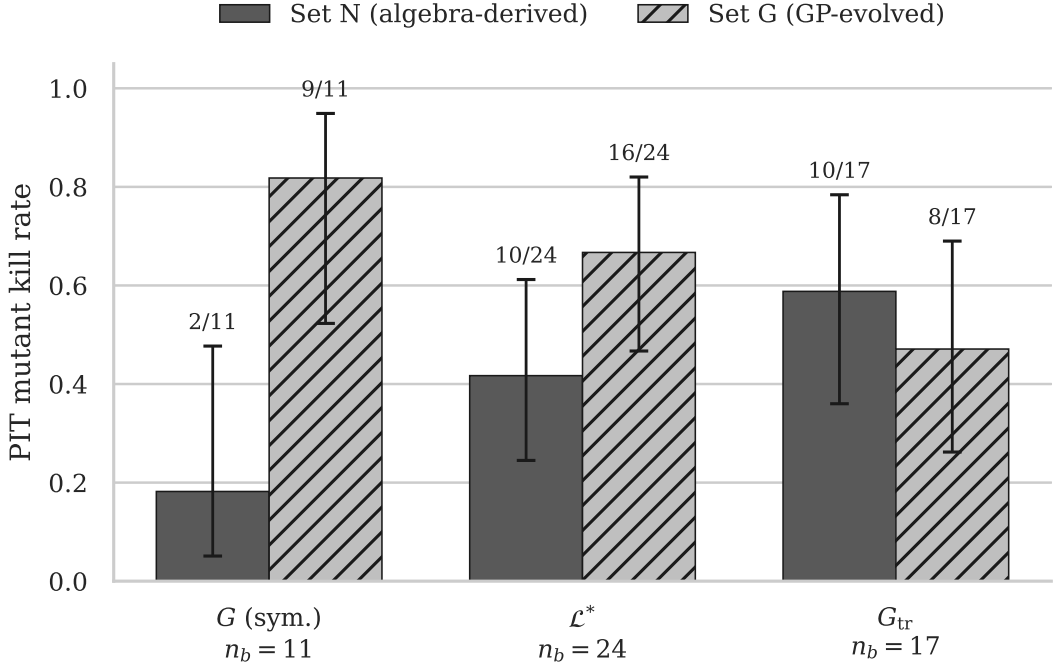


Fig. 6. Per-MetaPattern PIT kill rates show Set N and Set G as complementary rather than uniformly superior.

The MetaPattern set is *compositional* with respect to the underlying algebra: add a structural component, gain MetaPatterns; remove a structural component, lose them. The framework supports both construction and *prediction* of which patterns a new specialisation will exhibit, before any MR is identified empirically.

C Operator-algebraic definitions

This appendix gives the formal definitions of the operator-algebraic constructs summarised in §3.1.

C.0.1 Programs and program-induced operator algebras. Throughout the paper we treat a program P as a (possibly partial) computable function $P : \mathcal{X} \rightarrow \mathcal{Y}$. The class of programs of interest belongs to a *program family* $\mathcal{F}$ whose members share an underlying mathematical scaffolding.

DEFINITION 14 (PROGRAM-INDUCED OPERATOR ALGEBRA). Let P belong to a program family $\mathcal{F}$. A program-induced operator algebra of $\mathcal{F}$ is a tuple

$$\mathcal{A}_{\mathcal{F}} = (\mathcal{O}, \circ, \sim_{\mathcal{F}}),$$

where $\mathcal{O}$ is a set of operators acting on $\mathcal{X}$, $\mathcal{Y}$, or both; $\circ$ is an operator composition; and $\sim_{\mathcal{F}}$ is an equivalence relation declaring two operator expressions equal whenever they agree on every program of $\mathcal{F}$.

C.0.2 Symmetry groups (MetaPattern G).

DEFINITION 15 (SYMMETRY GROUP OF $\mathcal{A}_P$). A symmetry group of $\mathcal{A}_P$ is a subgroup $G \leq \mathcal{A}_P$ whose elements act on $\mathcal{X}$ such that, for all $P \in \mathcal{F}$ and all $g \in G$,

$$P(g \cdot x) = \rho(g) \cdot P(x) \quad \forall x \in \mathcal{X},$$

where $\rho : G \rightarrow \text{End}(\mathcal{Y})$ is a (possibly trivial) representation of G on $\mathcal{Y}$.

Accessible example. Rotating an input point cloud by any angle leaves a well-constructed shape classifier's predicted label unchanged, and rotates a predicted pose by the same angle.

C.0.3 Order operators: monotonicity and linearity (MetaPattern $O_{\leq}$).

DEFINITION 16 (MONOTONE OPERATOR). P is monotone with respect to θ if $\theta_1 \leq \theta_2 \Rightarrow P(\theta_1) \leq_{\mathcal{Y}} P(\theta_2)$ (or the reversed inequality, in which case P is anti-monotone).

DEFINITION 17 (LINEAR OPERATOR). P is linear on $X_0 \subseteq X$ when $P(\alpha x_1 + \beta x_2) = \alpha P(x_1) + \beta P(x_2)$ for all $x_1, x_2 \in X_0$ and scalars α, β .

Accessible example. Increasing a reactor's absorber cross-section can only lower, never raise, the computed multiplication factor k_{eff} ; the output moves monotonically with the input.

C.0.4 Self-adjoint operators (MetaPattern T^*).

DEFINITION 18 (SELF-ADJOINT OPERATOR). Given an inner product $\langle \cdot, \cdot \rangle$ on X , an operator $L \in O$ is self-adjoint if $\langle Lx, y \rangle = \langle x, Ly \rangle$ for all x, y in the domain of L .

Self-adjointness encodes a duality between the two arguments of an inner product. Reciprocity theorems in physics, transposed-graph identities in algorithms, and detailed-balance conditions in stochastic processes are all instances. In reactor physics, the adjoint transport formulation is self-adjoint under the appropriate inner product and gives source-detector reciprocity.

Accessible example. Swapping a source and a detector leaves the detector's reading unchanged.

C.0.5 Time-reversal operators (MetaPattern $\mathcal{T}_{\text{rev}}^*$).

DEFINITION 19 (TIME-REVERSAL OPERATOR). When X admits a time coordinate, a time-reversal operator $\mathcal{T}$ acts on inputs by reversing the time variable, $\mathcal{T}(x(t)) = x(-t)$. A program P is time-reversal symmetric on a sub-family of inputs when $P(\mathcal{T}x)$ is determined by $P(x)$ through a fixed bijection on $\mathcal{Y}$.

Time-reversal applies only where the underlying dynamics admit a reversible sub-family. Dissipative dynamics break this symmetry, so the corresponding MetaPattern is empty for those systems.

Accessible example. Integrating a frictionless (reversible) trajectory forward in time and then backward returns the system to its starting state up to round-off.

C.0.6 Limit operators (MetaPattern $\mathcal{L}^*$).

DEFINITION 20 (LIMIT OPERATOR). A parametrised limit operator $\mathcal{L}_{\theta}$ is a family of operators indexed by a parameter θ such that there exists a limit element $\mathcal{L}_*$ with $\mathcal{L}_{\theta} \rightarrow \mathcal{L}_*$ as $\theta \rightarrow \theta_*$ in an appropriate operator topology.

Accessible example. Halving the mesh spacing makes a convergent discretisation's output approach the same continuum solution.

C.0.7 Qualitative-dynamics operators (refinement of $O_{\leq}$).

DEFINITION 21 (QUALITATIVE-DYNAMICS OPERATOR). A qualitative-dynamics operator $\mathcal{D}$ is an operator on solution trajectories of an underlying ODE/PDE that extracts qualitative features such as extrema, inflection points, monotonic phases, overshoot magnitudes, S-curve transitions, and phase-portrait orbits, all of which are invariant under perturbations preserving the underlying dynamical structure (Sturm-type comparison theorems, dynamical-systems classification).

Some MR-relevant invariants are not point-wise but *shape-wise*: a solution curve may have an overshoot, a single extremum, or a monotone-then-saturating profile. Qualitative dynamics gives these qualitative-dynamics MRs their own algebraic root.

C.0.8 Method-comparison operators (refinement of $\mathcal{L}^*$).

DEFINITION 22 (METHOD-COMPARISON OPERATOR). A method-comparison operator $\mathcal{E}$ encodes a partial order $\preceq_{\mathcal{E}}$ on numerical or algorithmic methods, where $M_1 \preceq_{\mathcal{E}} M_2$ asserts that method M_1 produces an approximation no worse than method M_2 in a specified error norm, under specified conditions.

Appendices A, B (worked-example material)

Material running CONSTRUCT-MP on the heat / continuity / momentum / slowing-down reactor equations (Appendix A), per-MR source provenance for the representative reactor MRs (Appendix B), and worked examples for multi-structural-component-derivable MRs (originally introductory to Appendix C) is provided in supplementary S9_migrated_appendices/A_B_C_worked.tex; the body text's claims do not depend on this material beyond the cross-references made in §3.4 and §3.4.3.

D Detailed empirical tables

This appendix collects detailed empirical tables relocated from the main body for length conformance: the PIT-mutator/structural-component compatibility map, the $\mathcal{L}^*$ -blindness per-SUT kill rates, the cross-pipeline rediscovery on midpoint, and the per-MetaPattern head-to-head complementarity discussed in §5.3.6; the per-SUT kill-count table is in supplementary S9 (detailed_tables).

Table 11. PIT mutator categories are classified by whether they preserve each NOETHER structural component.

PIT mutator category	G	$O_{\leq}$	T^*	$\mathcal{T}_{\text{rev}}^*$	$\mathcal{L}^*$	$\mathcal{D}^*$	$\mathcal{E}^*$	$\mathcal{B}_{\text{rel}}^*$
CONDITIONALS_BOUNDARY	○	×	○	○	○	~	○	○
INCREMENTS	×	×	×	×	×	~	○	○
INVERT_NEGS	~	×	~	×	~	~	○	○
MATH (op swap)	×	~	×	×	○	×	~	○
NEGATE_CONDITIONALS	×	×	~	×	~	×	~	×
RETURN_VALS (zero/one)	○	○	×	×	○	○	~	○
VOID_METHOD_CALLS	○	○	○	○	○	×	~	○

E Proofs

Lemma C.1 (Well-foundedness of canonical MetaPattern ordering)

LEMMA 2. The canonical MetaPattern ordering of Definition 6 is a strict total order on the eight structural components, and the assignment of any multi-structural-component-derivable MR to its highest-priority structural component is unique.

PROOF. The ordering is a strict total order on a finite set by inspection. Given an MR ρ with derivations $\{(s_i, t_i)\}_{i=1}^k$ from structural components $s_1, \dots, s_k$, the assignment selects $s^* = \max_{>} \{s_1, \dots, s_k\}$. Since the strict order is total, s^* is unique. $\square$

Table 12. Per-SUT L_{scale} kill rates test the $\mathcal{L}^*$ -blindness prediction.

SUT	L_{scale} kill / mutants	Verdict
midpoint	0 / 3	Confirmed
clamp	0 / 7	Confirmed
signum	0 / 6 [†]	Confirmed (no L_{scale} kill)
gcdSig	0 / 9	Confirmed
lcmSig	0 / 11	Confirmed
hypotSig	2 / 4	Outlier; explained below
pooled (all six)	2 / 44	4.5%

[†] For signum the $\mathcal{L}^*$ -MetaPattern MR is positive-scalar invariance ($\text{signum}(\lambda x) = \text{signum}(x)$ for $\lambda > 0$) rather than degree-1 homogeneity; the prediction applies to this MR shape *a fortiori* since the right-hand side does not depend on λ .

Table 13. Cross-pipeline rediscovery on midpoint pairs GP-evolved MRs with algebra-derived counterparts.

GP-evolved MR (Set G)	Set N counterpart	MetaPattern
SwitchParams??1@2	G_swap (commutativity)	G
NumericAddition?1.000000?1	T_shift (translation by 1)	G _{tr}
NumericMultiplication?0.500000?2	L_scale (scaling, approx.)	$\mathcal{L}^*$

Table 14. The per-MetaPattern head-to-head decomposes Set N and Set G kills by operative structural component.

MetaPattern	n_b	Set N kills	rate (95% CI)	Set G kills	rate (95% CI)	both	N-only	G-only	neither
G (sym.)	11	2	0.182 [0.051, 0.477]	9	0.818 [0.523, 0.949]	2	0	7	2
$\mathcal{L}^*$	24	10	0.417 [0.245, 0.612]	16	0.667 [0.467, 0.820]	8	2	8	6
G _{tr}	17	10	0.588 [0.360, 0.784]	8	0.471 [0.262, 0.690]	7	3	1	6
D1 aggregate, PIT-covered MetaPatterns (secondary)	52	26	0.500 [0.369, 0.631]	37	0.712 [0.577, 0.817]	22	4	15	11
unmapped (lower-bound caveat)	25	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a

Per-structural-component instantiations of Translate

Definition 4 fixes the signature of Translate but defers per-structural-component specifics. Table 15 fills in the canonical input-tuple-generation rule and the resulting MR template for each of the eight structural components.

The per-structural-component rule together with the $\sim_s$ equivalence relation of Definition 3 fully determines $\text{Translate}(\iota, s)$ once the operator family $\Phi \subseteq s$ is fixed. Implementations of these seven cases as Python callables ship in supplementary material S1 (construct_mp.py, function translate together with the seven default_extractor_* routines).

Theorem 1 (Algebraic Closure under Translate), full proof

PROOF. *Existence.* Let $\rho \in \text{MR}(\mathcal{A}_P)$ in the sense of Definition 5. There exist a structural component s , an invariant $\iota \in \mathcal{I}_s$, and a derivation $\rho = \text{Translate}(\iota, s)$. Step 1 of CONSTRUCT-MP places ι in $\mathcal{I}_s$; step 2 places ρ in $\mathcal{R}(\iota)$; step 3 forms $m_s = \mathcal{R}(\iota) / \sim_s$; step 4 returns $m_s \in \mathbb{M}(\mathcal{A}_P)$.

Uniqueness. Suppose ρ admits derivations through multiple structural components: $\rho = \text{Translate}(\iota_1, s_1) = \text{Translate}(\iota_2, s_2)$ with $s_1 \neq s_2$. Definition 6's canonical MetaPattern ordering selects $s^* = \max_{>} \{s_1, s_2\}$. By Lemma C.1, s^* is unique; the canonical assignment $\rho \mapsto m_{s^*}$ is unique. $\square$

Table 15. Per-structural-component instantiations of Translate.

Structural component s	Canonical tuple from base x_0	MR template $\rho_{t,s}$
G	$(x_i)_{i=0}^{ G -1} = (g_i \cdot x_0)$ over group orbit	$\forall x_0, \forall g \in G : P(g \cdot x_0) = \rho(g) \cdot P(x_0)$
$O_{\leq}$	(x_1, x_2) with $x_1 \leq_{\theta} x_2$ in the partial order	$x_1 \leq_{\theta} x_2 \Rightarrow P(x_1) \leq_y P(x_2)$
T^*	paired tuple $((x_1, P(x_1)), (x_2, P(x_2)))$ in the inner product	$\langle LP(x_1), P(x_2) \rangle = \langle P(x_1), LP(x_2) \rangle$
$\mathcal{T}_{\text{rev}}^*$	$(x, \mathcal{T}x)$ with $\mathcal{T}$ the time-reversal involution	$P(\mathcal{T}x) = \mathcal{T}P(x)$ on the reversibility sub-domain
$\mathcal{L}^*$	sequence (x_{θ}) with $\theta \rightarrow \theta_*$	$\ P_{\theta} - P_{\theta_*}\ _* = O(f(\theta))$ at the prescribed rate
$\mathcal{D}^*$	solution trajectory $(\xi(t))_{t \geq 0}$ extracted by $\mathcal{D}$	qualitative-feature relation (extremum, monotonicity, S-curve) preserved on $(\xi(t))$
$\mathcal{E}^*$	method pair (M_1, M_2) in the partial order $\preceq_{\mathcal{E}}$	$\text{err}(M_1) \leq \text{err}(M_2)$ on the prescribed benchmark family
$\mathcal{B}_{\text{rel}}^*$	rewrite pair (E, E') generated by an algebraic-rewriting rule $\mathcal{R} \in \mathcal{R}_{\text{rel}}$ on the idempotent semiring	$\forall D. \text{eval}(E, D) \models_{\text{bag}} \text{eval}(E', D)$ at the rewriting rule's stated scope

Theorem 2 (Complexity), full proof

PROOF. *Step 1 cost.* For each generator g_i , compute g_i 's contribution to each $\mathcal{I}_s$. Cost: $O(t_i)$ per generator, $O(n \cdot \max_i t_i)$ aggregated.

Step 2 cost. ‘Translate’ is $O(1)$ per invariant. Total: $O(n)$.

Step 3 cost. Union-find with path compression: amortised $O(\alpha(n))$ per operation; we use $O(\log n)$ for simpler analysis. Total: $O(n \log n)$.

Step 4 cost. $O(7) = O(1)$.

Total: $O(n \cdot \max_i t_i \cdot \log n)$ when $\max_i t_i \geq 1$. $\square$

C.4 An open problem: absolute completeness

Theorem 1's scope is bounded by Definition 5: the closure is over MRs reachable through Translate from a single component invariant. We attempted a stronger statement:

CONJECTURE (THEOREM 1', ABSOLUTE COMPLETENESS). *Every MR ρ formulable as a property over $\mathcal{A}_P$'s operators is contained in some $m \in \mathbb{M}(\mathcal{A}_P)$.*

A proof would require either (a) a normalisation theorem reducing every operator-algebra MR to a Translate-reachable form, or (b) an extension of Translate to compositional invariants spanning multiple structural components. We were unable to establish either without imposing additional structural assumptions on $\mathcal{A}_P$. A partial empirical step toward Theorem 1' is available on a single domain through the supplementary-MR engineering audit reported in §3.4.3: 17 of 18 are placed by majority of three LLM labellers into one of the five MetaPatterns, the two refining families, or the relational extension $\mathcal{B}_{\text{rel}}^*$ (subsumption 94.4%, Fleiss' $\kappa = 0.857$, raw labels in supplementary S2 18mr_audit/); the single orphan localises to the metric-stability class discussed in Appendix 5 and Appendix C.5.2, for which a concrete candidate additional MetaPattern is sketched. This is a constructive partial support, not a proof: it is empirical, single-domain, bounded by the LLM-shared-training-data caveat noted in §3.4.3, and does not bound the gap in the absence of the candidate additional MetaPattern. The conjecture remains open. Section C.5 below documents three concrete classes of MRs that lie outside Theorem 1's scope and would have to be addressed

by any positive resolution of Theorem 1'; §3.7 and Appendix E additionally falsify Theorem 1' on $\mathcal{A}_{\text{PWR}}$.

C.5 Out-of-scope MRs: three concrete classes

We document three families of MRs that satisfy the informal description “a property of P 's executions over the operator algebra $\mathcal{A}_P$ ” but are not algebra-induced under Definition 5, and therefore lie outside the scope of Theorem 1. These are not threats to the theorem; they delimit it.

C.5.1 Probabilistic / distributional MRs without operator-algebraic representation. Consider the MR “the Shannon entropy $H(f(\mathbf{x}))$ of a classifier's output distribution should not decrease when the input is augmented by Gaussian noise $\mathcal{N}(0, \sigma^2 I)$ ”. The augmentation is parametric in σ and is not a group action, a partial-order operator, a self-adjoint operator, a time-reversal involution, a method-comparison operator, or (without further structure) a limit operator on $\mathcal{X}$. The MR constrains a functional H of the output distribution, not a point-wise relation between outputs. There is no $\iota \in \mathcal{I}_s$ for any s from which Translate can reach this MR. Consequently the MR is not algebra-induced under Definition 5 and is not covered by Theorem 1. A positive resolution of Theorem 1' would require either embedding stochastic perturbations into a probabilistic extension of $\mathcal{A}_P$, or extending Translate to functionals over output distributions.

C.5.2 Adversarial / input-set MRs. The MR “ $\|\delta\|_p \leq \epsilon \Rightarrow f(\mathbf{x} + \delta) = f(\mathbf{x})$ ” (adversarial robustness in ℓ_p ball) constrains f 's behaviour on a set defined by a norm constraint, not by a group action. The set $\{\delta : \|\delta\|_p \leq \epsilon\}$ does not in general carry a closed group structure on $\mathcal{X}$, and the MR is not derivable as $\text{Translate}(\iota, s)$ for any s in $\mathcal{D}(\mathcal{A}_P)$ as currently defined. Any treatment of this MR within NOETHER's scope would require an additional MetaPattern (e.g. a metric-ball MetaPattern with bounded Lipschitz operators), or a relaxation of Translate to non-group input neighbourhoods. A concrete candidate for the additional MetaPattern alluded to above is a metric-stability MetaPattern $M_{\text{lip}} = (\mathcal{X}, d_X) \rightarrow (\mathcal{Y}, d_Y)$ whose operators are K -Lipschitz maps and whose induced MetaPattern m_{lip} is the equivalence class of pointwise stability inequalities $d_Y(P(x'), P(x)) \leq K \cdot d_X(x', x)$. The corresponding Translate template constructs a follow-up input by metric perturbation $x' = x + \epsilon u$ (norm $\|u\| = 1$, $|\epsilon| = \delta$) and predicates a K -Lipschitz output bound; canonical MetaPattern ordering would place M_{lip} after $\mathcal{B}_{\text{rel}}^*$ since metric structure is independent of the eight existing component invariants (no group action on the perturbation set, no partial order on inputs, no self-adjoint operator, and no parametric refinement family). Theorem 1's closure proof transfers without modification because m_{lip} is single-MetaPattern algebraically derived. We do not commit to M_{lip} as part of the canonical decomposition in this paper, but we record it as the most concrete sub-instance of Remark (iv) (topological invariants) that admits an immediate Translate template.

C.5.3 Compositional MRs across multiple structural components under non-canonical derivations. The MR “rotating a point cloud by R and simultaneously taking the training-size limit $n \rightarrow \infty$ should give an output that converges to $f(\mathbf{x})$ at rate $O(1/\sqrt{n})$ ” invokes both G (rotation) and $\mathcal{L}^*$ (training-size limit) within a single MR. Definition 6's canonical MetaPattern ordering assigns this to G , but the resulting MR “ $f(R \cdot \mathbf{x}) = f(\mathbf{x})$ ” loses the convergence-rate content. The compositional content is in $\mathbb{M}(\mathcal{A}_P)$ if and only if it can be split into separate single-component invariants (one in G , one in $\mathcal{L}^*$); MRs whose semantic content is irreducibly compositional (in the sense that splitting destroys the property's truth value) lie outside Theorem 1's scope. This is the case Theorem 1' option (b) is targeted at: an extended Translate that operates on tuples of component invariants. We have not constructed such an extension that preserves Theorem 1's polynomial-time constructibility (Theorem 2), and we leave this as the most concrete sub-problem under Theorem 1'.

The three classes of §C.5.1–§C.5.3 are *abstract* characterisations of MRs outside Theorem 1’s scope. Appendix E supplements them with two *concrete instances* drawn from the PWR core diffusion algebra $\mathcal{A}_{\text{PWR}}$, each of which is empirically realised on every conforming PWR core simulator and documented in the standard reactor-physics literature. Together with §3.7, Appendix E establishes that Theorem 1’ (Conjecture E) is false on $\mathcal{A}_{\text{PWR}}$, and identifies five structural obstructions in Translate’s signature (operator-spectrum output, homomorphism-failure π -template, configuration-indexed adjoint structure, higher-order mixed-difference templates, bidirectional joint parametric dependence) that any positive resolution would have to repair.

C.6 Proofs for the negative instantiation on $\mathcal{A}_{\text{PWR}}$ (§3.7)

This appendix supplies the proofs for Propositions 1 and 2 of §3.7. The proof structure for both is the same: enumerate the eight structural components of $\mathcal{D}(\mathcal{A}_{\text{PWR}})$, instantiate the per-structural-component Translate template of Table 15, and verify by inspection that no invariant $\iota \in \mathcal{I}_s$ produces the target MR. The component-by-component exclusions in Proposition 1’s proof are the most detailed; Proposition 2 reuses the same exclusion pattern with the obstruction localised to the $O_{\leq}$ MetaPattern.

C.6.1 Proof of Proposition 1 (non-additivity is not Translate-reachable). Statement. For ρ_{nonadd} as in Definition 12 and every $s \in \mathcal{D}(\mathcal{A}_{\text{PWR}})$, every $\iota \in \mathcal{I}_s$: $\text{Translate}(\iota, s) \neq \rho_{\text{nonadd}}$.

Proof. We use the exact-form definition of $d\rho$ (Definition 11) throughout, which depends only on dominant eigenvalues k_{eff} of operators in $\mathcal{A}_{\text{PWR}}$. The adjoint-perturbation reading of §3.7 is the physical motivation but is not invoked in any case below.

Case $s = G$. By Definition 3, an invariant $\iota \in \mathcal{I}_G$ has the form (Φ, π) with $\Phi \subseteq G$ a finite operator family and π a relation on tuples $(x_i, P(x_i))_{i=1}^k$ obtained by applying Φ to a base input x_0 . By Table 15, the canonical Translate template for G is the equivariance schema

$$\text{Translate}(\iota, G) \equiv \forall x_0 \forall g \in \Phi : P(g \cdot x_0) = \rho(g) \cdot P(x_0).$$

Two structural mismatches with ρ_{nonadd} :

- Operator-spectrum output, not P output.* ρ_{nonadd} is an inequality between sums of $1/k_{\text{eff}}$ values, where each k_{eff} is the dominant eigenvalue of a configuration-specific diffusion operator $H_X \in \mathcal{A}_{\text{PWR}}$. The dominant eigenvalue is a *spectral* quantity of the operator, obtained as part of the simultaneous solution (k_{eff}, ϕ) of $H\phi = (1/k_{\text{eff}})F\phi$; it is not a function of $P(x)$ alone but a property of the operator H itself. Translate’s output relation π in Definition 3 ranges over $(X \times \mathcal{Y})^k$, where $\mathcal{Y}$ is the program output space (flux distributions, in this case). Operator-spectrum quantities are not in $\mathcal{Y}$; they are scalar invariants of operators in O , lying outside Translate’s signature by construction.
- Non-additivity is not equivariance.* Even granting a charitable extension to admit k_{eff} as a derived output, the G -template asserts equivariance of the output under the action of Φ : $P(g \cdot x_0)$ is determined by g and $P(x_0)$ through the representation $\rho(g)$. ρ_{nonadd} asserts that the worth functional $d\rho : O_{\text{rod}} \rightarrow \mathbb{R}_{>0}$ is *not a semigroup homomorphism*: $d\rho(A \cup B) \neq d\rho(A) + d\rho(B)$. Failure of homomorphism is not equivariance failure; it is the absence of an additive structure-preserving map, which has no expression in the equivariance template π for G . No $\iota \in \mathcal{I}_G$ produces ρ_{nonadd} .

Case $s = O_{\leq}$. By Table 15, the $O_{\leq}$ template is the absolute-monotonicity schema

$$\text{Translate}(\iota, O_{\leq}) \equiv \forall x_1, x_2 : x_1 \leq_{\theta} x_2 \implies P(x_1) \leq_{\mathcal{Y}} P(x_2).$$

ρ_{nonadd} is a *quaternary relation* on the four configurations $(x_0, O^A x_0, O^B x_0, O^{A \cup B} x_0)$: it asserts a non-vanishing mixed second difference

$$\begin{aligned} \Delta_{AB}(x_0) = & [k_{\text{eff}}^{-1}(P(x_0)) - k_{\text{eff}}^{-1}(P(O^{A \cup B} x_0))] \\ & - [k_{\text{eff}}^{-1}(P(x_0)) - k_{\text{eff}}^{-1}(P(O^A x_0))] \\ & - [k_{\text{eff}}^{-1}(P(x_0)) - k_{\text{eff}}^{-1}(P(O^B x_0))] \neq 0 \end{aligned}$$

(where the subtractions use Definition 11’s positive-worth convention). The $O_{\leq}$ template captures binary monotonicity between two points along a single partial order $\leq_{\theta}$. The mixed-difference structure of ρ_{nonadd} requires comparison across four configurations forming a “rectangle” $\{x_0, O^A x_0, O^B x_0, O^{A \cup B} x_0\}$ with two independent perturbation directions (insertion of A and insertion of B); no $\leq_{\theta}$ relates all four pairwise into a single chain. Furthermore, the assertion is *non-vanishing of a difference*, not a directional inequality; it is direction-agnostic, capturing both shadowing ($\Delta \downarrow 0$) and anti-shadowing ($\Delta \uparrow 0$). The $O_{\leq}$ template’s $\leq_{\mathcal{Y}}$ is a directional partial order, not a non-vanishing constraint. No $\iota \in \mathcal{I}_{O_{\leq}}$ produces ρ_{nonadd} .

Case $s = T^*$. By Table 15, the T^* template asserts $\langle L P(x_1), P(x_2) \rangle = \langle P(x_1), L P(x_2) \rangle$ for a single self-adjoint operator L in a fixed inner product. The structural mismatch with ρ_{nonadd} has two components:

- (i) *Single operator L vs. configuration-indexed family.* The exact $d\rho$ values entering ρ_{nonadd} are eigenvalues of *four distinct diffusion operators* $H_{\emptyset}, H_A, H_B, H_{A \cup B}$, each self-adjoint within its own configuration but pairwise distinct as operators. The T^* template’s L is fixed; it does not range over a configuration-indexed family $\{H_X\}_{X \in \mathcal{O}_{\text{rod}}}$. The self-adjointness of any single H_X does not imply or constrain a relation between eigenvalues of *different* H_X ’s.
- (ii) *Adjoint weighting function depends on configuration.* Under the standard adjoint-perturbation reading, each $d\rho(B; X)$ for $X \in \{\emptyset, A\}$ admits the first-order representation

$$d\rho(B; X) \approx - \frac{\langle \phi_X^{\dagger}, \delta H_B \phi_X \rangle}{\langle \phi_X^{\dagger}, F_X \phi_X \rangle},$$

with $(\phi_X, \phi_X^{\dagger})$ the principal eigenfunctions of $(H_X, H_X^{\dagger})$. The Hilbert-space inner product $\langle \cdot, \cdot \rangle = \int (\cdot)(\cdot) d\Omega dE d\mathbf{r}$ is unchanged across X . What changes across X is the *adjoint weighting function* $\phi_X^{\dagger}$, which is the principal eigenfunction of a structurally different adjoint operator $H_X^{\dagger}$ from $H_{\emptyset}^{\dagger}$ (because H_X contains A ’s absorbing material whereas $H_{\emptyset}$ does not). $\phi_A^{\dagger}$ is locally depressed in A ’s geometric support and globally redistributed elsewhere; this is the adjoint-perturbation root cause of the non-additivity $d\rho(A \cup B) \neq d\rho(A) + d\rho(B)$.

The T^* template’s structure (single L , fixed inner product, asserting $\langle Lx_1, x_2 \rangle = \langle x_1, Lx_2 \rangle$) has no place to express “the weighting function $\phi^{\dagger}$ entering the inner product is itself the principal eigenfunction of a configuration-indexed adjoint operator and changes when configuration changes”. This is a configuration-dependent adjoint structure, not a self-adjointness identity on a single operator. No $\iota \in \mathcal{I}_{T^*}$ produces ρ_{nonadd} .

Case $s = \mathcal{T}_{\text{rev}}^*$. The PWR diffusion operator $-\nabla \cdot D \nabla + \Sigma_a$ is irreversible: it is parabolic (transient form) or elliptic (steady-state form), neither of which admits a time-reversal involution on the relevant solution sub-family. Hence $\mathcal{T}_{\text{rev}}^*$ is empty for $\mathcal{A}_{\text{PWR}}$, and vacuously no $\iota \in \mathcal{I}_{\mathcal{T}_{\text{rev}}^*}$.

Case $s = \mathcal{L}^*$. By Table 15, the $\mathcal{L}^*$ template is a convergence statement $\|P_{\theta} - P_{\theta_*}\|_* = O(f(\theta))$ at a parametric limit. ρ_{nonadd} contains no limit operation: it is a strict inequality at finite, fixed configurations $(\emptyset, A, B, A \cup B, x_0)$. No $\iota \in \mathcal{I}_{\mathcal{L}^*}$.

Case $s = \mathcal{D}^*$. By Table 15, the $\mathcal{D}^*$ template asserts a qualitative-feature relation (extremum, monotonicity, S-curve) on a solution trajectory $\xi(t)$. ρ_{nonadd} is a steady-state inequality between four reactivity values; it does not concern trajectory shapes. No $\iota \in \mathcal{I}_{\mathcal{D}^*}$.

Case $s = \mathcal{E}^*$. By Table 15, the $\mathcal{E}^*$ template compares two *methods* M_1, M_2 on a benchmark family. ρ_{nonadd} compares four *operator configurations* on a single fixed method (the same diffusion solver P in all four worth values). No $\iota \in \mathcal{I}_{\mathcal{E}^*}$.

Case $s = \mathcal{B}_{\text{rel}}^*$. Per §3.6, $\mathcal{B}_{\text{rel}}^*$ is non-empty only on program families with idempotent-semiring rewriting structure. The PWR diffusion solution operator algebra does not carry such structure (no rewriting rules between core states preserve evaluation under all valid inputs). $\mathcal{B}_{\text{rel}}^*$ is empty for $\mathcal{A}_{\text{PWR}}$. Vacuously no $\iota \in \mathcal{I}_{\mathcal{B}_{\text{rel}}^*}$.

This exhausts $\mathcal{D}(\mathcal{A}_{\text{PWR}})$. For every structural component s and every $\iota \in \mathcal{I}_s$, $\text{Translate}(\iota, s) \neq \rho_{\text{nonadd}}$. Hence $\rho_{\text{nonadd}} \notin \text{MR}(\mathcal{A}_{\text{PWR}})$. $\square$

C.6.2 Three obstructions identified by Proposition 1's proof. The proof identifies three independent structural obstructions in Translate 's present definition:

- (O1) *Operator-spectrum output is not in $\mathcal{Y}$.* ρ_{nonadd} asserts a relation between $1/k_{\text{eff}}$ values, where each k_{eff} is a dominant eigenvalue of an operator H_X . Translate 's π in Definition 3 ranges over $(\mathcal{X} \times \mathcal{Y})^k$; eigenvalues of operators in $\mathcal{O}$ are scalar invariants of those operators, not elements of $\mathcal{Y}$.
- (O2) *Output relation is non-additivity (failure of homomorphism), not equivariance, partial order, or self-adjointness.* The worth functional $d\rho : \mathcal{O}_{\text{rod}} \rightarrow \mathbb{R}$ is not a semigroup homomorphism; this is a third type of algebraic relation distinct from the equivariance, monotonicity, and self-adjointness expressed by Translate 's per-structural-component π templates.
- (O3) *Adjoint weighting function $\phi_X^\dagger$ entering the T^* MetaPattern's inner product is configuration-dependent.* While the Hilbert-space measure $d\Omega dE dr$ is fixed, the eigenfunction $\phi_X^\dagger$ varies with X because $H_X^\dagger$ varies with X . Definition 3 fixes the self-adjoint operator L once; it does not admit a configuration-indexed family $\{L_X\}$ with L_X 's spectrum varying with X .

A constructive resolution of Proposition 1's obstructions would require Translate to admit (i) operator-spectrum output relations (eigenvalues, integrals, ratios as the targets of π), (ii) homomorphism-failure relations as a π -template type alongside equivariance / monotonicity / self-adjointness, and (iii) configuration-dependent adjoint structure on T^* .

C.6.3 Proof of Proposition 2 (MTC-vs-boron mixed dependence is not Translate -reachable). Statement. For $\rho_{\text{MTC-bor}}$ as in Definition 13 and every $s \in \mathcal{D}(\mathcal{A}_{\text{PWR}})$, every $\iota \in \mathcal{I}_s$: $\text{Translate}(\iota, s) \neq \rho_{\text{MTC-bor}}$.

Proof. The principal obstruction is in the $\mathcal{O}_{\leq}$ MetaPattern; we treat that case in detail and abbreviate the others.

Case $s = \mathcal{O}_{\leq}$. By Table 15, the $\mathcal{O}_{\leq}$ template is the absolute-monotonicity schema

$$\text{Translate}(\iota, \mathcal{O}_{\leq}) \equiv \forall x_1, x_2 : x_1 \leq_{\theta} x_2 \implies P(x_1) \leq_{\mathcal{Y}} P(x_2),$$

a *first-order* statement asserting a binary relation between $P(x_1)$ and $P(x_2)$ at θ -comparable inputs along a single partial-order direction $\leq_{\theta}$. $\rho_{\text{MTC-bor}}$ asserts a *non-zero second-order mixed partial derivative* of k_{eff} with respect to two *independent* parameter directions T_{mod} and C_B :

$$\left| \frac{\partial^2 k_{\text{eff}}}{\partial T_{\text{mod}} \partial C_B} \right| \neq \tau_{\text{MTC-bor}}.$$

Two independent obstructions in the $\mathcal{O}_{\leq}$ template:

- (a) *Order vs. mixed-derivative structure.* The mixed second derivative is the limit of a four-point finite-difference quotient over a “rectangle” $\{(T_0, C_0), (T_0 + \Delta T, C_0), (T_0, C_0 + \Delta C), (T_0 + \Delta T, C_0 + \Delta C)\}$.

$\Delta C)\}$:

$$\frac{\partial^2 k_{\text{eff}}}{\partial T_{\text{mod}} \partial C_B} = \lim_{\Delta T, \Delta C \rightarrow 0} \frac{k(T_0 + \Delta T, C_0 + \Delta C) - k(T_0, C_0 + \Delta C) - k(T_0 + \Delta T, C_0) + k(T_0, C_0)}{\Delta T \cdot \Delta C}$$

(writing k for k_{eff}). This is structurally a *four-point relation*, not a two-point relation. The $O_{\leq}$ template's π relates $(P(x_1), P(x_2))$ pairwise; it has no expression for a four-point combination weighted by $1/(\Delta T \cdot \Delta C)$. Equivalently, the canonical input-tuple-generation rule for $O_{\leq}$ in Table 15 produces tuples (x_1, x_2) with $x_1 \leq_{\theta} x_2$, not 2-by-2 perturbation rectangles.

- (b) *Two independent parameter directions*. The mixed derivative requires *jointly varying* T_{mod} and C_B along two independent directions. The $O_{\leq}$ template's $\leq_{\theta}$ is a *single* partial order on $\mathcal{X}$ (or on a single coordinate of $\mathcal{X}$). Even granting the construction of a product order $\leq_T \times \leq_C$ on a 2-D parameter slice of $\mathcal{X}$, the π relation still applies along the order chain as a directional inequality, not as a non-vanishing-second-difference statement. The two independent parameter directions are perpendicular, not chained; the $\leq_{\theta}$ formalism collapses them into one chain only by losing the non-vanishing-mixed-difference content.

Furthermore, $\rho_{\text{MTC-bor}}$'s output is again k_{eff} , an operator-spectrum quantity (cf. Proposition 1 Case G obstruction (a)). This gives a *third* obstruction in $O_{\leq}$: even if mixed-second-derivative structure could be embedded into π , the output value would not lie in $\mathcal{Y}$. No $\iota \in \mathcal{I}_{O_{\leq}}$ produces $\rho_{\text{MTC-bor}}$.

Other structural components.

- *Case $s = G$* . No group action in $\mathcal{A}_{\text{PWR}}$ relates (T_{mod}, C_B) pairs at different parameter values to each other through equivariance: T_{mod} and C_B are continuous parameters acting on the cross-section library, not group-action coordinates. Even if a charitable embedding in G were attempted, the same operator-spectrum-output obstruction (Proposition 1 Case G (a)) applies: k_{eff} is not in $\mathcal{Y}$.
 - *Case $s = T^*$* . The MTC operator $\partial/\partial T_{\text{mod}}$ is not self-adjoint in any natural inner product on the diffusion-solution space (the parameter T_{mod} enters the cross-section coefficients of H rather than H 's acting space, so $\partial H/\partial T_{\text{mod}}$ has no self-adjointness structure analogous to H itself). No $\iota \in \mathcal{I}_{T^*}$.
 - *Case $s = \mathcal{T}_{\text{rev}}^*$* . Empty for PWR diffusion. No $\iota \in \mathcal{I}_{\mathcal{T}_{\text{rev}}^*}$.
 - *Case $s = \mathcal{L}^*$* . $\rho_{\text{MTC-bor}}$ does not assert convergence at a parametric limit; it asserts a non-zero finite value of a mixed second derivative at a *finite* parameter point in the operating envelope. The mixed-derivative limit $\Delta T, \Delta C \rightarrow 0$ is the *definition* of the derivative, not the MR's claim; the MR's claim is that the resulting derivative exceeds $\tau_{\text{MTC-bor}}$, which is a non-vanishing condition at a finite parameter value, not a convergence rate. No $\iota \in \mathcal{I}_{\mathcal{L}^*}$.
 - *Case $s = \mathcal{D}^*$* . $\rho_{\text{MTC-bor}}$ concerns k_{eff} as a function of two parameters at steady state; it does not involve a solution trajectory of an underlying ODE/PDE. No $\iota \in \mathcal{I}_{\mathcal{D}^*}$.
 - *Case $s = \mathcal{E}^*$* . The comparison is between two parameter regimes (T_0, C_0) and $(T_0 + \Delta T, C_0 + \Delta C)$ of a single fixed method (the same PWR core simulator P in all four eigenvalues), not between two methods. No $\iota \in \mathcal{I}_{\mathcal{E}^*}$.
 - *Case $s = \mathcal{B}_{\text{rel}}^*$* . Empty for $\mathcal{A}_{\text{PWR}}$. No $\iota \in \mathcal{I}_{\mathcal{B}_{\text{rel}}^*}$.
- This exhausts $\mathcal{D}(\mathcal{A}_{\text{PWR}})$. Hence $\rho_{\text{MTC-bor}} \notin \text{MR}(\mathcal{A}_{\text{PWR}})$. $\square$

C.6.4 Two further obstructions identified by Proposition 2's proof. The proof identifies two further independent structural obstructions in Translate's present definition, in addition to the three identified by Proposition 1:

- (O4) *MR is a non-zero second-order mixed partial derivative, not a first-order relation*. All per-structural-component π templates in Table 15 are first-order: equivariance is a first-order

identity $P(gx_0) = \rho(g)P(x_0)$; monotonicity is a first-order inequality $P(x_1) \leq P(x_2)$; self-adjointness is a first-order pairing $\langle Lx_1, x_2 \rangle = \langle x_1, Lx_2 \rangle$. Mixed second differences (and a fortiori higher-order mixed differences) have no expression in any of these.

(O5) *MR involves two independent parameter directions, joined as a 2-by-2 perturbation rectangle, not as a chain.* The $O_{\leq}$ MetaPattern's $\leq_{\theta}$ is a single partial-order direction; even with multiple independent partial orders in $I_{O_{\leq}}$, the π template relates pairwise inputs along a *single chosen direction*, not jointly across two perpendicular directions in a finite-difference rectangle.

A constructive resolution of Proposition 2's obstructions would require Translate to admit (iv) higher-order mixed-difference π -templates and (v) bidirectional joint parametric dependence beyond the single- θ partial order of Definition 3.

C.6.5 Combined corollary.

COROLLARY (THEOREM 1' IS FALSE ON $\mathcal{A}_{\text{PWR}}$, TWO-FOLD). *The MRs ρ_{nonadd} , $\rho_{\text{MTC-bor}}$ are each formulable over operators of $\mathcal{A}_{\text{PWR}}$ and each empirically realised on every conforming PWR core simulator. By Propositions 1 and 2, neither is in $\text{MR}(\mathcal{A}_{\text{PWR}})$. The structural obstructions O1–O5 identified by the two proofs are pairwise distinct: no single extension of Translate's signature absorbs any two simultaneously, so the joint obstruction is irreducibly five-fold.*

REMARK 13 (OPEN: COMPOSITE Translate). *A natural follow-up is to define a Composite Translate $\text{Translate} : \mathcal{I}_{s_1} \times \cdots \times \mathcal{I}_{s_k} \rightarrow \text{MR}(P)$ that combines invariants from multiple structural components under a generalised π template admitting (i) operator-spectrum output, (ii) homomorphism-failure relations, (iii) configuration-indexed adjoint structure, (iv) higher-order mixed differences, and (v) bidirectional joint parametric dependence. Whether such an extension preserves Theorem 1's closure (now over $\text{MR}(\mathcal{A}_P)$) and Theorem 2's polynomial-time constructibility is the principal open problem this subsection leaves to future work. Five independent extensions are needed; a single uniform Composite Translate covering all of them would be a substantive theoretical contribution.*

C.7 Worked enumeration of CONSTRUCT-MP on the Boltzmann algebra (migrated)

The line-by-line enumeration of CONSTRUCT-MP Steps 1–4 on $\mathcal{A}_{\text{Boltz}}$, originally Appendix C.7, is provided in supplementary S9_migrated_appendices/C7_boltzmann_worked.tex.

Appendices D, E (implementation + consistency check)

A reference implementation of CONSTRUCT-MP (Appendix D) and the construct-trace consistency check on hand-crafted structural-component-targeted mutants (Appendix E) are provided in supplementary S9_migrated_appendices/D_E_implementation_consistency.tex; both are confidence-strengthening illustrative material that does not carry inferential weight for the paper's RQ4 verdicts.

Data Availability Statement

The companion artefact for this paper, including the CONSTRUCT-MP implementation, the structural decompositions for the three case-study algebras, the Set N MR catalogue (§5.3.3: 36 MRs on MathSignalClass + ComplexSignal; ISSUE-009: 5 MRs on Apache Commons Math), the Set G GenMorph rerun output, the LLM-ensemble Set L harvest with token-cost logs, the PIT 1.7.4 mutation traces, and the per-MetaPattern + D1/D2 aggregation pipeline, is archived on Zenodo under DOI [10.5281/zenodo.20250634](https://doi.org/10.5281/zenodo.20250634). Reproduction scripts (setup.sh, run_all.sh, tests/run.sh) and the experiment-side CLAUDE.md collaboration record are bundled. The camera-ready version will record any newer Zenodo version DOI if the replication package is updated during review.